

STRASBORG-ENTZHEIM, France

WMO: 071900

Lat: 48.5494N

Lon: 7.6403E

Elev: 505

StdP: 14.43

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.5	22.1	12.2	10.4	19.4	16.8	13.1	25.0	26.7	49.9	23.3	47.6	5.1	350	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	89.5	70.3	85.9	68.5	82.6	67.4	71.8	85.7	70.1	82.7	68.6	80.0	6.4	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	102.3	77.4	65.8	97.0	75.5	64.4	92.4	73.9	35.8	86.1	34.3	82.8	33.1	80.2	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	17.6	15.3	12.2	95.0	5.8	3.3	8.0	97.3	4.6	99.2	1.3	101.0	-2.9	103.4		
			11.5	75.1	5.8	1.5	7.3	76.2	3.9	77.1	0.7	77.9	-3.5	78.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	36.1	38.5	45.0	52.1	59.4	65.5	68.4	67.7	60.5	52.6	43.3	37.7	
	DBStd	13.33	8.17	7.41	6.82	6.78	6.42	6.43	5.83	5.89	5.50	6.52	6.64	7.71	
	HDD50	1650	434	325	181	56	5	0	0	0	1	49	215	384	
	HDD65	5023	897	742	620	389	196	72	32	37	157	386	650	845	
	CDD50	2497	3	3	26	118	297	466	570	550	315	130	16	3	
	CDD65	394	0	0	0	1	23	88	137	122	21	2	0	0	
	CDH74	4058	0	0	0	57	291	895	1409	1176	215	15	0	0	
	CDH80	1318	0	0	0	4	62	281	509	419	42	1	0	0	
Wind	WSAvg	6.4	7.1	7.4	7.7	6.8	6.6	6.2	6.0	5.3	5.7	5.6	5.8	7.0	
Precipitation	PrecAvg	24.6	1.4	1.3	1.5	1.7	3.0	2.8	2.6	2.6	2.2	2.0	1.9	1.7	
	PrecMax	32.0	4.1	3.1	4.5	5.6	8.0	7.2	8.0	5.3	4.9	5.2	4.1	4.8	
	PrecMin	15.7	0.1	0.2	0.2	0.4	0.7	0.6	0.7	0.2	0.5	0.1	0.1	0.2	
	PrecStd	4.0	0.8	0.7	0.9	1.1	1.5	1.3	1.3	1.2	1.1	1.2	1.0	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	60.6	70.2	79.8	86.2	92.6	94.2	95.1	85.2	76.0	63.5	57.0	
		MCWB	51.9	50.5	53.8	60.0	66.4	71.4	70.8	71.6	67.1	63.2	56.1	51.4	
	2%	DB	53.7	56.0	65.1	74.8	80.9	87.1	89.5	88.6	79.8	70.2	58.9	54.0	
		MCWB	49.1	48.9	52.2	57.7	63.6	69.0	70.5	70.3	65.8	60.8	53.3	50.0	
	5%	DB	50.5	52.7	60.4	69.9	77.0	83.0	85.7	84.4	75.8	66.2	55.8	51.3	
		MCWB	46.7	47.0	50.7	55.5	61.7	67.9	68.4	68.5	63.9	58.9	51.8	47.9	
	10%	DB	47.4	49.6	56.4	65.7	73.0	79.2	81.9	80.7	71.9	62.7	53.0	48.5	
		MCWB	44.3	45.0	49.1	53.7	60.3	65.8	67.3	67.1	61.5	57.0	49.8	45.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	51.8	56.2	61.9	68.9	73.7	73.7	73.6	69.2	64.1	57.0	52.7
			MCDB	56.2	58.2	65.7	76.4	81.9	87.7	88.5	88.2	80.1	72.6	61.7	55.7
2%		WB	49.5	49.5	53.6	58.8	65.8	71.0	71.7	71.2	67.1	61.8	54.3	50.3	
		MCDB	53.0	55.3	61.3	70.7	77.3	82.8	86.1	85.3	77.6	68.5	58.0	53.4	
5%		WB	46.9	47.4	51.7	56.7	63.4	69.0	70.0	69.7	65.0	59.9	52.2	48.0	
		MCDB	50.3	52.2	58.9	67.7	73.6	80.5	82.7	81.9	73.8	65.0	55.5	51.0	
10%		WB	44.4	45.1	49.7	54.7	61.3	67.1	68.4	68.0	62.9	58.0	50.0	46.0	
		MCDB	47.0	49.0	56.2	64.1	70.3	77.6	79.4	78.8	70.0	62.0	52.7	48.5	
Mean Daily Temperature Range	5% DB	MDBR	9.4	12.8	16.7	19.7	19.4	20.4	21.0	21.0	19.5	16.2	10.7	9.0	
		MCDBR	12.9	18.9	25.3	28.2	27.1	26.6	28.5	28.3	25.9	21.6	15.1	12.3	
		MCWBR	9.8	12.7	14.4	13.7	12.4	11.9	12.1	11.9	12.9	13.0	10.7	9.3	
	5% WB	MCDBR	12.3	17.4	20.9	25.2	23.3	24.5	25.2	25.5	23.2	20.0	14.2	11.6	
		MCWBR	9.7	12.0	12.6	12.8	11.4	11.8	11.3	11.6	12.1	12.6	10.5	9.1	
Clear-Sky Solar Irradiance	taub	0.317	0.340	0.370	0.404	0.414	0.420	0.416	0.412	0.387	0.366	0.336	0.313		
	taud	2.449	2.385	2.319	2.248	2.260	2.278	2.276	2.308	2.353	2.405	2.443	2.441		
	Ebn at Noon	238	257	266	268	269	268	267	262	257	243	227	223		
	Edh at Noon	21	27	34	40	41	41	40	37	32	26	21	18		
All-Sky Solar Radiation	RadAvg	294	532	919	1359	1607	1846	1719	1494	1136	678	339	236		
	RadStd	45	93	141	187	178	153	168	153	117	56	36	40		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (82 neighbors)		N/A	N/A	N/A	N/A	+0.59	+0.50	N/A	N/A	N/A	N/A

Nomenclature: See separate page